Extending the Financial Industry Business Ontology to Represent Time-Varying Exchange-Traded Fund Allocations

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Abstract

Exchange-traded fund (ETF) providers publish sector and geographic allocation percentages, but differences in identifiers, dates and classifications make this information difficult to combine and trace. We review regulatory representations, provider disclosures, commercial databases and semantic models, including the Financial Industry Business Ontology (FIBO). Based on this comparison, we propose a draft conceptual model connecting each dated ETF allocation disclosure to its percentage entries, categories, classification context and captured source. The model separates a customer's portfolio holdings from the allocations reported by fund providers. Our proposed contribution is a lightweight FIBO application pattern for preserving this disclosure context and supporting later relational and semantic representations.

1 Introduction

Consider a bank customer who transfers savings into an investment account and purchases ETFs. An ETF is an investment fund whose units trade on a stock exchange and which holds assets such as company shares. The customer's investments form a portfolio. Although the bank records which ETFs the customer owns, an analyst may also need to explain the industries and countries represented within those funds.

Providers publish allocation summaries showing the percentage associated with each sector or geographic category. However, "30% technology" needs its fund, observation date, classification context and source to be understood correctly. Similar category labels or different reporting dates can make percentages unsuitable for direct comparison. A customer's holding date may also differ from the provider's allocation date.

FIBO supplies shared financial concepts [3], and earlier FIBO-based work demonstrates dated fund reporting and allocation summaries [2]. We investigate a small application pattern that explicitly connects an observed ETF allocation disclosure to its entries and supporting context. The project uses one or two fictional portfolios, two or three ETFs, and public Vanguard and BlackRock summaries [1, 11]. Investment recommendations, machine learning and capstone risk analysis remain outside the scope.

2 Brief History

Fund information was recorded in paper reports and tables before computerized reporting. U.S. disclosure requirements developed under the Investment Company Act of 1940 [6], while EDGAR moved filings online by 1996 [5]. Later representations included XBRL-tagged prospectus summaries [8] and structured XML filings: N-PORT reports position-level holdings and N-CEN provides an annual fund census [9]. FIBO introduced shared semantics, while the Global Fund Reporting Ontology (GFRO) demonstrated point-in-time fund graphs and country, sector and currency summaries [2, 3]. Commercial databases aggregate fund information for cross-fund analysis [4]. These U.S. regulatory examples provide historical background; our sample disclosures concern Canadian ETFs.

3 Hierarchy of Representations

Figure 1 groups representations by their purpose and structure. Regulatory models support oversight through narrative reports, XML schedules and tagged prospectus summaries. Provider disclosures and commercial aggregators support access to fund information and comparisons across funds. Semantic representations support shared meaning and interoperability through FIBO and applications such as GFRO. These categories overlap in use, but distinguish the modelling choices relevant to our disclosure pattern.

4 Comparative Analysis

Table 1 compares what the representations preserve and how they inform our design. Regulatory schedules organize fund–security positions, whereas provider sector and geographic summaries organize category percentages. We therefore distinguish a customer's dated ETF *Holding* from a provider's *AllocationEntry*. Daily security-level holdings under U.S. Rule 6c-11 are also distinct from the aggregate summaries we model [10].

Commercial classifications help comparisons, but category meaning should remain connected to its scheme and provider context. FIBO contributes reusable financial semantics; fund-type classifications and investment-policy limitations should not be confused with categories in an observed allocation. GFRO already demonstrates dated reporting, so our contribution concerns an explicit integration pattern rather than the invention of dates or weights. The reviewed material does not explicitly describe our particular combination of provider disclosure, entries, classification context and captured source [2, 3].

5 Project Positioning and Draft ERD

Figure 2 separates the customer investment mix from each ETF's disclosed allocation. *Holding* connects one *Portfolio* to one ETF and records its weight and holding date. *ETFIdentifier* preserves the identifier scheme and exchange context where needed.

The main refinement is *AllocationSnapshot*: one ETF, one allocation date, one dimension and one captured source. A snapshot contains one or more *AllocationEntry* records, each connecting a category to its disclosed fraction. New observations create new snapshots. *SourceDocument* retains the URL, retrieval time and captured file; publication time is optional. *ClassificationScheme* is optional when a provider does not identify it. Unstated schemes

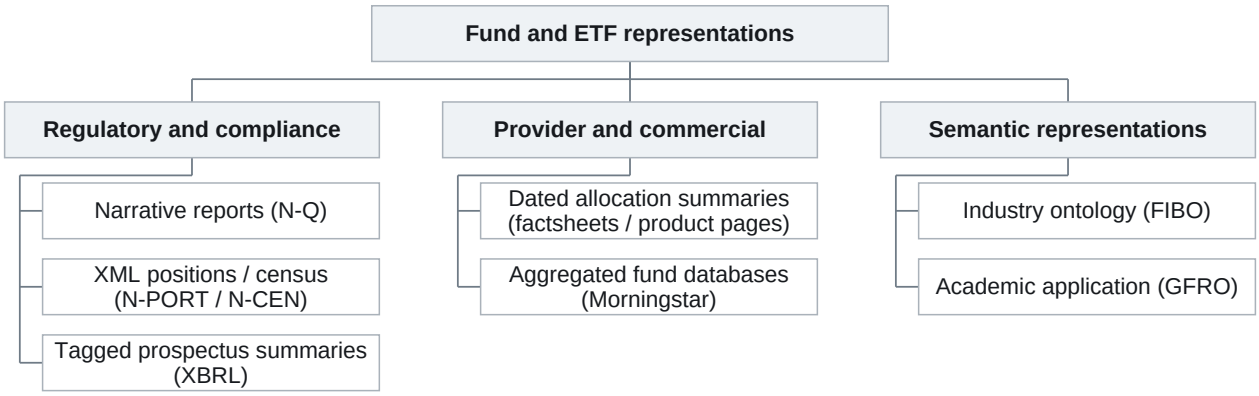


Figure 1: Representations of fund and ETF information, grouped by purpose and then by format or application. Regulatory examples are U.S. background; provider allocation summaries are the project's data sources.

Table 1: Comparison of representations and implications for the ERD.

Representation	Structure and strengths	Limitation and design response
Narrative reports [7]	Dated fund–security holdings in human-readable tables.	Extraction is needed; separate a disclosure from its entries.
Regulatory XML [9]	N-PORT: fund–position schedules; N-CEN: annual census.	Regulatory purpose differs; retain structure and observation dates.
Provider summaries [1, 11]	ETF–date–category percentages in factsheets or product pages.	Meaning and coverage vary; preserve source, dimension and basis.
Commercial data [4]	Normalized fund data and category classifications.	Retain classification context and source links in our model.
FIBO [3]	Fund types, policies, weighted structures and date semantics.	Reuse cautiously; distinguish policy from disclosure observation.
GFRO [2]	Point-in-time graphs with country, sector and currency summaries.	Acknowledge overlap; make our disclosure integration explicit.

remain unknown, and categories are not merged merely because labels match.

Entries must match their snapshot's dimension and consistent classification context when known. Basis and coverage notes preserve exclusions and partial disclosures. Portfolio exposure estimates combine holding weights with compatible category fractions and retain the selected dates and sources; missing categories are not silently treated as zero. The model's six competency questions cover holdings, sector and geographic exposure, source traceability, ETF identity, and missing or outdated observations. Later SQL and SPARQL versions will be evaluated against the same hand-checked cases. This milestone proposes the model; implementation and evaluation remain future work.

Repository: https://github.com/Pyrex111/CSC501_Data_Modeling_Project

References

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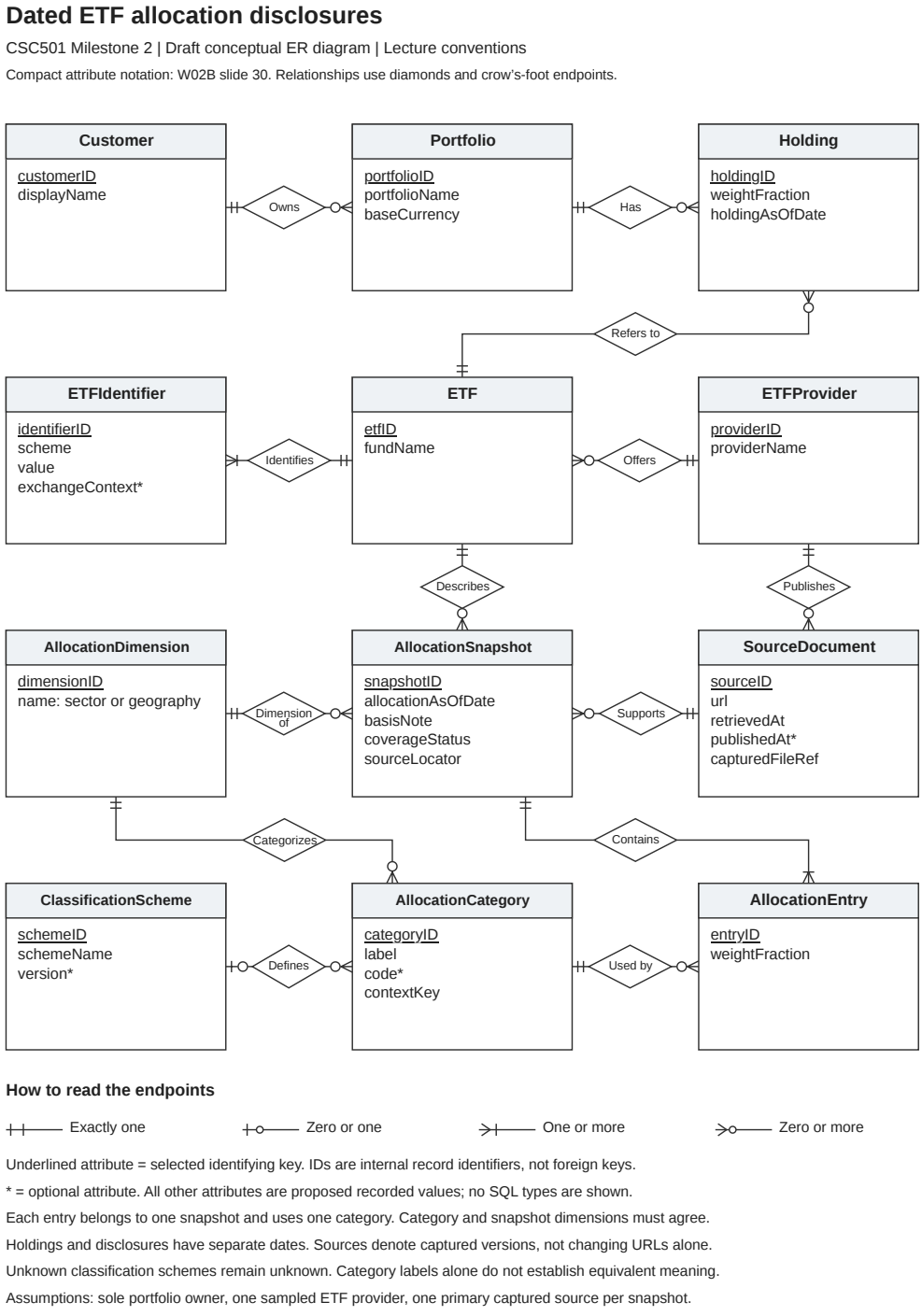


Figure 2: Draft conceptual ER diagram for dated ETF allocation disclosures. Holding dates and provider allocation dates are separate. Each entry belongs to one snapshot and uses one category; dimensions and known scheme context must agree. Cardinalities reflect the small course dataset and are not final OWL restrictions.

ETF allocation model contribution

Taimoor | CSC501 Milestone 2 | 5 October 2026

This contribution contains the draft ER diagram explanation and the design notes for our team review. The report text below can be combined with Mason and Shahmeer's sections. The separate one-page ERD is intended for the diagram page; the supporting notes are not all part of the three-page main report.

Draft text for the report

Our draft ER diagram starts with a customer and their investment portfolio, which is the collection of investments they own. A Holding connects a Portfolio to an ETF, a fund traded on a stock exchange, and records its portfolio weight and observation date. This separates the customer's investment mix from the allocation reported for each fund. ETFIdentifier also stores the identifier scheme and, where needed, exchange context, so we do not rely on a ticker alone to identify an ETF.

The main refinement is AllocationSnapshot. Each snapshot describes one ETF, one date and one allocation dimension, either sector or geography. It contains AllocationEntry records, each connecting a category to its disclosed percentage. A new observation creates a new snapshot rather than replacing the previous percentages. SourceDocument records the captured provider document or webpage, its URL and retrieval time. ClassificationScheme gives categories their stated classification context, when the source provides it. Keeping the allocation basis and coverage notes also helps us interpret disclosures that exclude cash or show only part of a fund [4, 5].

FIBO provides relevant building blocks, but their meaning needs care. Its fund-classification concepts describe the type of fund, and definesAllocations links investment policy to allocation limitations [1]. We therefore represent a published allocation as an observation, rather than as a policy. FIBO's weighted-basket and date concepts are also candidates for reuse, subject to their ontology restrictions [2, 3]. Browne et al. already demonstrate FIBO-based reporting of dated fund information and allocation summaries [6]. Our proposed contribution is narrower: an explicit pattern connecting a provider-published ETF allocation to its date, entries, categories, classification context and source. This is a proposed integration gap, not a claim that FIBO lacks weights or dates.

The model supports portfolio exposure estimates by combining holding weights with compatible disclosed category weights. The result should retain the selected snapshot dates and sources. We will evaluate the later relational and semantic versions with the same competency questions, using one or two fictional portfolios and two or three ETFs. Full holdings reconstruction, return prediction and portfolio optimisation remain outside this project.

Suggested figure caption

Draft conceptual ER diagram for dated ETF allocation disclosures. Holding records the customer's ETF weight at a portfolio observation date; AllocationSnapshot records a provider's disclosure date. The disclosure pattern connects an ETF, source, dimension, entries and category context. Cardinalities describe the small course dataset and are not final OWL restrictions.

Entity meanings and relationships

The diagram is conceptual. Each box shows a record identifier and the attributes needed for our questions. Foreign keys, implementation types and the final OWL class mappings come later. The relationship list also fixes how to read the endpoint cardinalities.

Relationship	Cardinality and meaning
Customer to Portfolio	One customer owns zero or more portfolios; each portfolio has one owner. Sole ownership is enough for the fictional examples.
Portfolio and ETF through Holding	Each holding belongs to one portfolio and refers to one ETF; either can have zero or more holdings. A holding is one dated observation of an ETF's portfolio weight, not a trade.
ETF to ETFIdentifier	Each ETF has one or more identifiers; each identifier record identifies one ETF. Scheme, value and required exchange context determine identity.
ETFProvider to ETF	A provider offers zero or more ETFs; each sampled ETF has one provider. Use the verified provider name. Do not infer the legal issuer or investment manager from the brand.
ETF to AllocationSnapshot	Each ETF has zero or more snapshots; each snapshot describes one ETF. Sector and geography observations are separate snapshots.
AllocationDimension to AllocationSnapshot	Each snapshot has one dimension; a dimension has zero or more snapshots. The two dimension values are sector and geography.
AllocationSnapshot to AllocationEntry	Each accepted snapshot contains one or more entries; each entry belongs to one snapshot. An entry stores one disclosed category percentage.
AllocationCategory to AllocationEntry	Each entry uses one category; a category can be reused in zero or more entries. Reused categories must have the same meaning and classification context.
AllocationDimension to AllocationCategory	Each category has one dimension; a dimension has zero or more categories. Every entry's category dimension must match its snapshot dimension.
ClassificationScheme to AllocationCategory	A scheme defines zero or more categories; a category has zero or one known scheme. A scheme record includes its version when available. An unknown scheme remains unknown.
ETFProvider to SourceDocument	A provider publishes zero or more captured sources; each source has one publisher in this dataset. The publishing role is independent of the provider relationship to the ETF.
SourceDocument to AllocationSnapshot	A source supports zero or more snapshots; each snapshot has one primary source. A source means a captured version. One factsheet can support sector and geography snapshots.

Why these entities are included

Snapshot and Entry separate a disclosure from its percentage rows. Categories, schemes and dimensions preserve their meaning. Sources preserve the evidence. Customers and portfolios support the fictional examples; a bank or savings account can introduce the story without needing its own entity for these questions.

Proposed modeling rules

Observation grain. A Holding is unique for portfolio, ETF and holdingAsOfDate. An AllocationSnapshot is one source-backed observation for ETF, allocationAsOfDate, dimension and allocation basis. Two sources for the same fund and date remain separate observations; never average them silently. Within a snapshot, a category appears at most once.

Dates and source versions. allocationAsOfDate is the date of the reported allocation. retrievedAt is when we captured the source; publishedAt is optional and only recorded if stated. Holding dates are separate. Keep a captured file reference, page or section locator, and optionally a content hash. A changing URL alone does not preserve an earlier disclosure.

Weight values and totals. Store percentages as decimal fractions from 0 to 1, while preserving displayed precision in the captured evidence. Check the sum of the complete ETF-only portfolio for each holding date. For allocation snapshots, check totals only when the categories are mutually exclusive and cover the stated basis. Record partial or unknown coverage and provider rounding; do not force every disclosure to sum to exactly 1.

Categories and schemes. Do not merge categories only because their labels match. Known categories are identified within their scheme and version. If the scheme is unstated, keep categories local to the provider and disclosed allocation context. All entries in a snapshot use the same dimension and a consistent scheme context when known. Geography uses one level per snapshot, such as country or region, so overlapping levels are not added together.

Selecting disclosures. For a portfolio dated D, select at most one compatible snapshot per ETF and dimension, with allocationAsOfDate no later than D. Prefer an exact date match; otherwise explicitly use the latest prior observation and report its age. A freshness threshold is a query parameter, not a universal rule. Distinct sources tied on date require an explicit selection or an ambiguity result. A missing snapshot is reported as missing.

Exposure calculation. For category c, multiply each selected holding weight by its ETF's disclosed category fraction, then sum the compatible contributions. Check dates, category meaning, classification context and allocation basis first. A missing category is zero only if a complete disclosure establishes this; otherwise it is unknown. Excluded cash or other assets make the result an estimate on the provider's stated basis, rather than an exact percentage of all portfolio assets.

Storing and implementing the model. Exposure and snapshot age are derived query results, so they are not extra core entities. Each result should report the selected holding observations, snapshot identifiers, dates and sources. These rules are proposed data constraints for later relational and semantic validation; a conceptual ERD does not enforce them by itself. No trades, prices, underlying-security positions, bank accounts, analyst workflow or ML entities are needed for the submitted competency questions.

Competency questions for later validation

These six draft questions turn the scope in our proposal into checks for the later relational and semantic representations. They are a validation plan, not completed implementation results. Both versions should return the same selected observations and numerical answers under the same date and source-selection rules.

ID	Question	Relationships and expected result
CQ1	Which ETFs does a customer hold in a portfolio on date D, and at what weights?	Customer > Portfolio > Holding > ETF, filtered by holdingAsOfDate.
CQ2	What sector contribution or estimate can be calculated for that portfolio from compatible disclosures?	Holdings plus selected sector snapshots, entries and categories; return dates, coverage and missing data.
CQ3	What geographic contribution or estimate can be calculated on the same terms?	Use geographic snapshots at one compatible geography level; keep the provider's basis visible.
CQ4	Which provider source supports a selected ETF allocation percentage?	Entry > Snapshot > SourceDocument > ETFProvider, including source locator and retrieval time.
CQ5	Which ETF does a supplied identifier refer to?	ETFIdentifier > ETF, using scheme and any required exchange context; flag ambiguous input.
CQ6	Which ETF allocations are missing or older than a chosen threshold at D?	Compare D with selected allocationAsOfDate; distinguish stale observations from missing ones.

Review cases that can reveal model problems

Use a hand-checked exposure example, two dates for one ETF, a missing snapshot, an old snapshot, a ticker that needs exchange context, an unstated classification scheme, and two sources tied on the same observation date. Check that the output preserves the selected source and date, flags ambiguity, and does not convert missing values into zero. This gives us more useful validation than checking only that the tables or graph can be loaded.

Worked example within the proposal

Sara's fictional portfolio has CAD 20,000, with 45% in XIC and 55% in VFV. The Vanguard page reports a 100.00% United States market allocation dated 31 August 2026 [4]. VFV's contribution is 0.55 times 1.00, or 0.55. This becomes the total portfolio US estimate only if a compatible XIC disclosure establishes its contribution too. An unreported category is not automatically zero, and an equity-only disclosure does not give a complete valuation of every portfolio asset.

The VFV disclosure is 36 days old on 6 October 2026, so a query with a chosen 30-day threshold would flag it. That threshold is only an example. These are retrospective as-of queries: retrieving a source after the portfolio date does not prove that it was publicly available on that date.

How the sources inform the design

Source or representation	Relevant evidence	Design response
Provider disclosures [4, 5]	VFV presents a dated country allocation and source caveats about exposure coverage. XIC has a dated sector breakdown identified as a percentage of market value.	Separate dimension and as-of date; retain source, basis and coverage notes.
FIBO fund concepts [1]	FundClassification classifies a collective investment vehicle by fund type. definesAllocations has policy and limitation classes as its domain and range.	Do not type a sector-percentage row as a fund classification or use the policy property for an observation.
FIBO basket and date concepts [2, 3]	WeightedBasket and WeightedBasketConstituent support weighted structure. hasAsOfDate is an object property for the relevant observation date.	Investigate reuse of weights and dates. Check superclass and property restrictions before assigning OWL types.
GFRO application [6]	The paper reports point-in-time fund graphs and country, sector and currency summaries. It does not present enough of the disclosure schema to establish the absence of our exact pattern.	Acknowledge this overlap. Position our work as a small explicit disclosure pattern, with the gap still subject to the team's ontology review.

FIBO reuse decisions to check with Shahmeer

The definitions in references 1 to 3 were inspected directly in the RDF files at the same pinned revision. FundClassification and FundClassificationScheme classify a fund by asset class. Our AllocationCategory instead identifies a category used in an observed breakdown. Generic classifier and classification-scheme concepts are better candidates for that role; the exact imported Commons classes need to be checked before the semantic mapping is finalized.

WeightedBasket requires a weighted constituent and a weighting expression, and WeightedBasketConstituent specializes BasketConstituent. A disclosure is information about an ETF, so it should not automatically be typed as an investment basket. hasAsOfDate is an object property, not a direct date-valued datatype property. The later RDF mapping must represent its date value in the expected way. We can preserve the meaning of these concepts without claiming every ERD entity is already a FIBO class.

The separately examined Ownership module defines Portfolio as a collection of holdings, while Publishers defines Publisher as a party role [7, 8]. These are further reuse candidates. Our dated Holding record is an observation of a position, and our ETFProvider box records an organization, so neither is an automatic one-to-one OWL mapping. An organization may play a publisher role without being identical to that role.

The gap claim remains limited to an explicit integrated disclosure pattern. This review of selected modules and a prior application is not an exhaustive search of every FIBO import. Shahmeer should check neighboring reporting, classification and fund modules, and the current status of the proposed pattern. If an equivalent pattern exists, our contribution can be framed as a small specialization or application rather than claiming it is absent.

References and evidence record

Mason can renumber these references or use the included bibliography. Retain captured provider disclosures for the later dataset. The contribution pack includes editable diagram files and a compact FIBO evidence extract.

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Representations of ETF and Fund Allocation Information: A Historical Review and Comparative Analysis

A Historical Review:

Fund portfolio allocation data is a core information asset in global financial markets. Its recording and representation have evolved from unstructured paper narratives to standardized structured data and semantic models, driven by regulatory transparency mandates, market efficiency demands, and computer technology.

1. Brief History

In the past 90 years, fund allocation disclosure has evolved from unstructured narrative reports to standardized, machine-readable formats. Under U.S. regulation, the Investment Company Act of 1940 required registered funds to disclose portfolio holdings in periodic shareholder reports and prospectuses [7], and from 2004 in quarterly Form N-Q filings [8]; for decades these records were plain-text tables designed for human reading. The first U.S. ETF (SPDR S&P 500 ETF Trust, SPY, 1993) introduced daily portfolio transparency [1], formalized for U.S. ETFs in 2019 by SEC Rule 6c-11 [12], while the SEC's EDGAR system (fully phased in by 1996) moved filings online [6]. Regulators then added machine-readable formats: XBRL-tagged prospectus summaries in 2009 [9] and, in 2016, Forms N-PORT and N-CEN, which replaced text-based reports with XML filings — N-PORT reports monthly position-level holdings data filed quarterly, and N-CEN is an annual fund census [10, 11]. In parallel, the Financial Industry Business Ontology (FIBO) standardized fund and security semantics [2, 3], and Browne et al. demonstrated FIBO-based reporting of dated fund-level information and allocation summaries; commercial databases such as Morningstar aggregate these disclosures for cross-fund research [4, 5]. Because our sample ETFs are Canadian, the U.S. regulatory examples serve as background: they establish the disclosure patterns — dated, structured, provider-published allocation percentages — that motivate our model.

2. Summary

Table 1 compares the representations most relevant to our model. Throughout, we keep two concepts distinct: Holding, a customer's investment in an ETF at a portfolio observation date, and AllocationEntry, a disclosed percentage (e.g., "technology: 30%") within a dated AllocationSnapshot published by the ETF provider.

Table 1 — Comparison of fund allocation representation paradigms

Representation	What It Captures	What It Leaves Out	What We Can Learn for Our Design	Related ERD Entities
Regulatory narrative filings (U.S.; N-SAR / N-Q [7, 8])	Audited holdings at reporting dates; the fund–security relationship	Unstructured; manual extraction; holdings not linked to allocation categories	Keep a disclosure separate from its entries; our model records disclosed percentages, not full holdings	ETF; AllocationSnapshot (dated disclosure)
Daily ETF website disclosures (Rule 6c-11 [12, 1])	Provider-published daily basket: securities, weights, publication date	No uniform schema; no category or classification links; source versions not tracked	Model each disclosure as AllocationSnapshot (ETF + date + dimension); preserve the source in SourceDocument	ETFProvider, AllocationSnapshot, AllocationEntry, AllocationDimension, SourceDocument
Structured regulatory filings (U.S.; N-PORT / N-CEN [10, 11])	N-PORT: standardized monthly position-level holdings; N-CEN: annual fund census	Regulatory schedules and a 60-day lag; not daily provider disclosures	Structured, dated disclosure is feasible; we model provider-published percentages, not filing schedules	ETF; AllocationSnapshot (structured, dated)
FIBO semantic ontology (incl. Browne et al. [2, 3])	Formal fund and security semantics; Browne et al.: dated fund information and allocation summaries	Conceptual only; no instance-level allocation entries	Reuse core semantics; add an explicit pattern linking ETF, date, entries, categories, classification context, source	ETF, ETFIdentifier, ClassificationScheme, AllocationCategory
Commercial aggregated databases (e.g., Morningstar [4, 5])	Normalized holdings and category data across funds	Proprietary classification logic; limited per-entry source and date traceability	Model ClassificationScheme as a separate entity and keep source links	ETFProvider, ClassificationScheme, AllocationCategory, SourceDocument

Summary: Implications for Our Model

The comparison informs our ERD in three ways:
(1) Holding (a customer's ETF weight at a holding date) and AllocationEntry (a disclosed percentage within a dated AllocationSnapshot) are modeled as distinct entities;
(2) each AllocationSnapshot links an ETF to an allocation date, dimension, entries, and — where the source provides it — a classification scheme, with SourceDocument preserving the captured source;
(3) existing representations cover parts of this pattern (e.g., Browne et al.'s FIBO-based reporting of dated fund information), so our contribution is an explicit integration pattern rather than a claim that prior work lacks dates or weights.

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[11] U.S. Securities and Exchange Commission. 2016. Investment Company Reporting Modernization Rules — Small Entity Compliance Guide. Available at https://www.sec.gov/divisions/investment/guidance/secg-investment-company-reporting-modernization-rules

[12] U.S. Securities and Exchange Commission. 2019. Exchange-Traded Funds (Final Rule; Rule 6c-11). Available at https://www.federalregister.gov/documents/2019/10/24/2019-21250/exchange-traded-funds

Hierarchy Text

Existing representations of fund and ETF allocation information can be organized by their structural paradigms and primary use cases into three distinct categories: Regulatory & Compliance Models, Operational & Commercial Aggregators, and Semantic & Knowledge Representations.

First, Regulatory & Compliance Models are designed for mandatory oversight and public accountability. This category has evolved from unstructured narrative documents (such as U.S. SEC Forms N-SAR and N-Q) into machine-readable formats, including structured XML filings for position holdings (Form N-PORT) and census reporting (Form N-CEN), alongside semantic XBRL taxonomies used for tagging prospectus summaries.

Second, Operational & Commercial Aggregators prioritize market mechanics and cross-fund analysis. This includes high-frequency dated ETF allocation disclosures—such as provider factsheets and product-page allocation summaries driven by U.S. SEC Rule 6c-11 to support pre-market operations—and commercial databases like Morningstar. These proprietary databases normalize heterogeneous holdings data into standard relational schemas for easier investor benchmarking.

Finally, Semantic & Knowledge Representations focus on formal, cross-system interoperability. This category includes standardized financial ontologies like the Financial Industry Business Ontology (FIBO) and academic semantic applications such as GFRO, which use RDF/OWL to define shared industry concepts.

While these models successfully support regulatory audits, daily trading, and high-level conceptual definitions, our project targets a specific opportunity. We propose an explicit semantic pattern that links a dated ETF allocation disclosure to its source provenance, underlying entries, and classification scheme.

Mermaid Vertical Graph

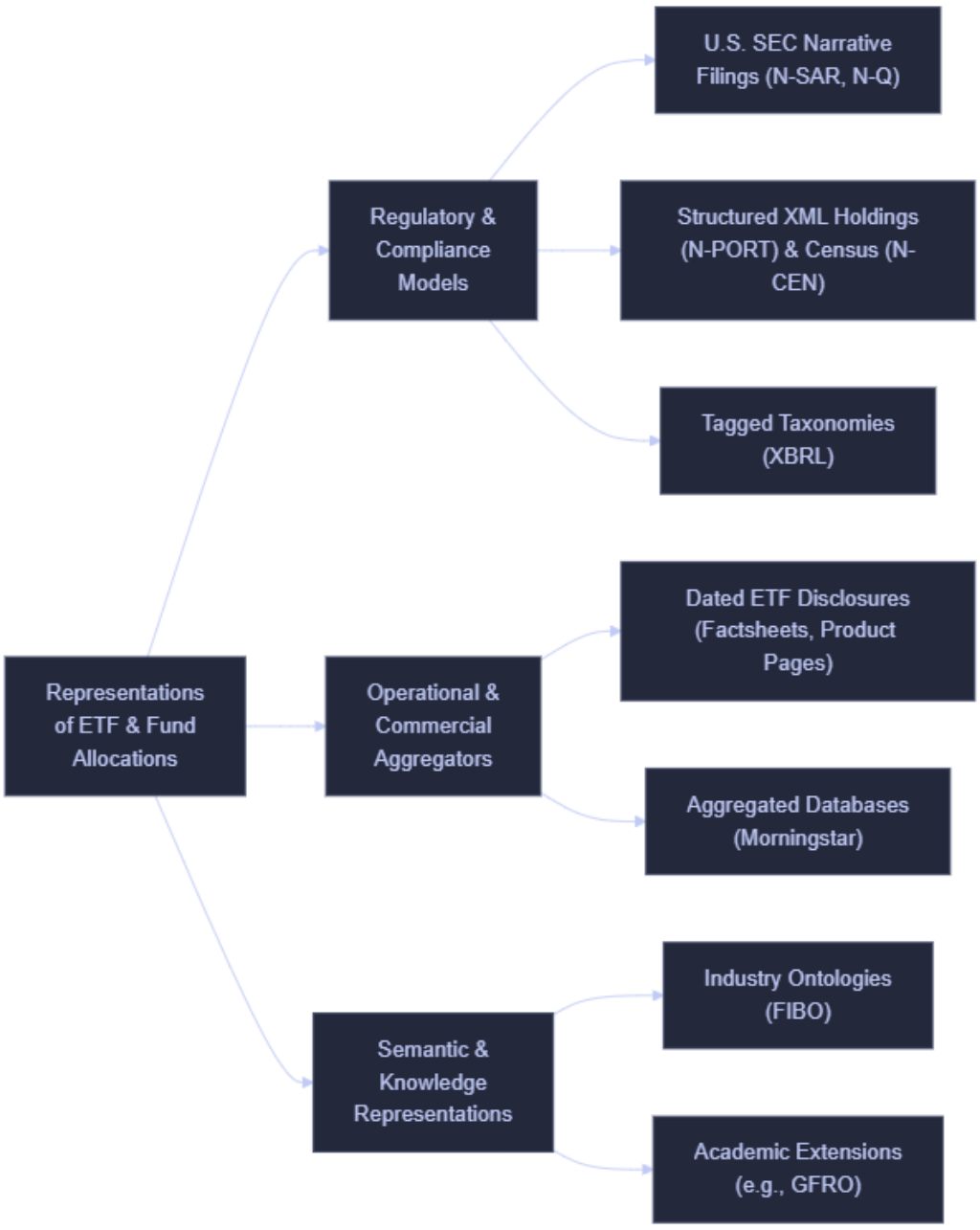
```
graph LR
    Root["Representations<br>of ETF & Fund<br>Allocations"]

    %% Regulatory Branch
    Root --> Reg["Regulatory &<br>Compliance<br>Models"]
    Reg --> Reg1["U.S. SEC Narrative<br>Filings (N-SAR, N-Q)"]
    Reg --> Reg2["Structured XML Holdings<br>(N-PORT) & Census (N-CEN)"]
    Reg --> Reg3["Tagged Taxonomies<br>(XBRL)"]

    %% Operational Branch
    Root --> Ops["Operational &<br>Commercial<br>Aggregators"]
    Ops --> Ops1["Dated ETF Disclosures<br>(Factsheets, Product Pages)"]
    Ops --> Ops2["Aggregated Databases<br>(Morningstar)"]

    %% Semantic Branch
    Root --> Sem["Semantic &<br>Knowledge<br>Representations"]
    Sem --> Sem1["Industry Ontologies<br>(FIBO)"]
    Sem --> Sem2["Academic Extensions<br>(e.g., GFRO)"]
```

Appendix A3 (AI-assisted editing): Original student draft supplied by Shahmeer



Mermaid Horizontal Graph

```
graph TD

  Root["Representations of ETF<br>& Fund Allocations"]


  %% Regulatory Branch

  Root --> Reg["Regulatory &<br>Compliance Models"]

  Reg --> Reg1["U.S. SEC Narrative Filings<br>(N-SAR, N-Q)"]

  Reg --> Reg2["Structured XML Holdings (N-PORT)<br>& Census (N-CEN)"]

  Reg --> Reg3["Tagged Taxonomies<br>(XBRL)"]


  %% Operational Branch

  Root --> Ops["Operational & Commercial<br>Aggregators"]

  Ops --> Ops1["Dated ETF Disclosures<br>(Factsheets, Product Pages)"]

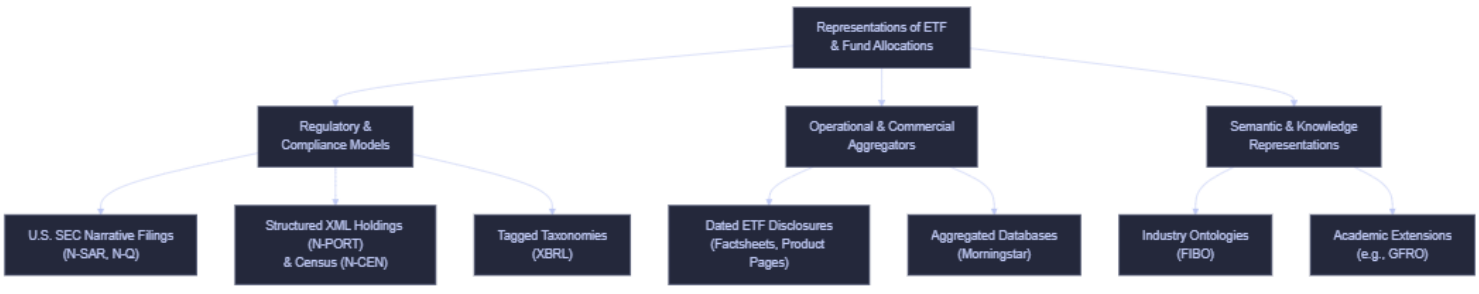
  Ops --> Ops2["Aggregated Databases<br>(Morningstar)"]


  %% Semantic Branch

  Root --> Sem["Semantic & Knowledge<br>Representations"]

  Sem --> Sem1["Industry Ontologies<br>(FIBO)"]

  Sem --> Sem2["Academic Extensions<br>(e.g., GFRO)"]
```



Gap Statement

The Financial Industry Business Ontology (FIBO) provides a standardized vocabulary for financial concepts like funds, portfolios, and reporting dates. While previous research, such as the GFRO application by Browne et al., successfully extends FIBO to support dated point-in-time fund reporting, the reviewed material does not describe our particular integrated pattern.

Specifically, there remains an opportunity to define a lightweight, reusable application pattern linking an ETF directly to its allocation date, percentage entries, categories, classification context, and supporting source document. Furthermore, this proposed model explicitly distinguishes an empirical allocation observation—what a provider actually reported on a given date from a fund's general investment policy, which defines limitations on what the fund is allowed to hold. This pattern addresses the gap by integrating ETF allocation snapshots while preserving their original source provenance.

FIBO Concept Verification & Gap Validation

Text for the Team / Appendix:

1. FIBO Modules for Reuse and Concept Verification

After auditing the relevant Financial Industry Business Ontology (FIBO) modules (inspected at the pinned October 1, 2026 revision), I have verified our team's proposed semantic mappings and reuse decisions:

- **Classification vs. Categories:** In the CollectiveInvestmentVehicles.rdf module, FundClassification and FundClassificationScheme classify an entire collective investment vehicle by its broad fund type (e.g., equity vs. fixed income). This confirms Taimoor’s finding: these concepts do not represent individual observed holding categories like "Technology" or "Canada" within a fund. We will need generic classifier concepts for our AllocationCategory entity.

- **Policy vs. Observation:** FIBO uses the property definesAllocations to link a FundPortfolioInvestmentPolicy to specific portfolio investment limitations. This defines a normative constraint on what a fund is permitted to hold. Therefore, our ER diagram correctly represents an ETF breakdown as a dated, empirical observation of a disclosure, rather than an investment policy constraint.
- **Baskets & Dates:** In the FinancialProductsAndServices.rdf module, the WeightedBasket concept requires a weighted constituent and a specific weighting expression. A disclosure is high-level information about an ETF, so it should not automatically be typed strictly as an investment basket. Furthermore, in FinancialDates.rdf, hasAsOfDate is an object property that links to a structured date individual, not a direct date-valued literal. Our later RDF mapping must handle this date structure explicitly.
- **Publishers & Portfolios:** FIBO's Ownership.rdf module defines Portfolio as a collection of holdings, while the Publishers.rdf module defines Publisher as a functional party role. Our model's ETFProvider entity records an organization, acknowledging that an organization may play a publishing role without being identical to the role itself.

2. Gap Validation Against Prior Literature To ensure our project advances the state of the art, I reviewed the 2017 application of FIBO by Browne et al., which developed the Global Fund Reporting Ontology (GFRO).

- **Prior Work Scope:** Browne et al. demonstrate point-in-time fund graphs and geographic/sector summaries using FIBO.
- **Proposed Integration Gap:** While GFRO covers dated fund reporting, the reviewed material does not explicitly detail a lightweight integration pattern linking a provider-published website disclosure (factsheet) directly to its captured source document, publication timestamp, underlying percentage entries, and classification context.
- **Positioning Our Contribution:** Our project does not claim that FIBO lacks the capability to represent weights, dates, or classifications. Instead, we propose a narrow, explicit application pattern that bridges this integration

gap, ensuring that published ETF allocations retain their source provenance and semantic meaning.

Citations

@inproceedings{browne2017semantic,

title={Semantic Ontologies and Financial Reporting: An Application of the FIBO},

author={Browne, Oliver and Krdzavac, Nenad and O'Reilly, Philip and Hutchinson, Mark},

booktitle={Proceedings of the Joint Ontology Workshops 2017},

volume={2050},

series={CEUR Workshop Proceedings},

year={2017}

}

@misc{fibo2026,

author={{EDM Council}},

title={Financial Industry Business Ontology (FIBO)},

howpublished={\url{https://spec.edmcouncil.org/fibo/}},

note={Pinned revision examined October 1, 2026},

year={2026}

}

Addition for Masons Comparison table.

Existing Model: GFRO (Browne et al.) / Prior Academic Work

Main Purpose: Point-in-time semantic fund reporting and regional summaries.

Key Entities: Fund; Geographic Region; Sector; Currency.

Key Relationships: Fund has regional/sector allocations at a point in time.

Strengths: Demonstrates practical FIBO extension for aggregated fund reporting.

Limitations: Lacks a lightweight pattern for daily ETF disclosures; omits source document provenance and classification scheme context.

Implications for ER Design: Validates the use of formal semantics but confirms the gap for an explicit source-attributed disclosure pattern.

Abstract

Exchange-traded fund (ETF) providers publish sector and geographic allocation percentages, but differences in identifiers, dates and classifications make this information difficult to combine and trace. We review regulatory representations, provider disclosures, commercial databases and semantic models, including the Financial Industry Business Ontology (FIBO). Based on this comparison, we propose a draft conceptual model connecting each dated ETF allocation disclosure to its percentage entries, categories, classification context and captured source. The model separates a customer's portfolio holdings from the allocations reported by fund providers. Our proposed contribution is a lightweight FIBO application pattern for preserving this disclosure context, providing a foundation for later relational and semantic representations.

1 Introduction

Consider a bank customer who transfers some savings into an investment account and purchases exchange-traded funds (ETFs). An ETF is an investment fund whose units trade on a stock exchange and which holds a collection of assets, such as company shares. The customer's investments form a portfolio. Although the bank records which ETFs the customer owns, an analyst may also need to explain the industries and countries represented within those funds.

ETF providers publish allocation summaries showing the percentage associated with each sector or geographic category. However, a value such as "30% technology" needs its fund, observation date, classification context and source to be understood correctly. Different reporting dates or similar category labels can make percentages unsuitable for direct comparison. The customer's holding date may also differ from the date of the provider's disclosure.

FIBO provides a shared vocabulary for financial concepts (EDM Council, 2026), and earlier FIBO-based research already demonstrates dated fund reporting and allocation summaries (Browne et al., 2017). Building on these representations, we investigate a small application pattern that explicitly connects an observed ETF allocation disclosure to its entries and supporting context. This milestone reviews existing representations and develops a draft ER diagram. The project uses one or two fictional portfolios, two or three ETFs, and public Vanguard and BlackRock

summaries. Later stages will compare relational and semantic representations through questions about exposure, identity, source traceability and outdated observations. Investment recommendations, machine learning and risk analysis remain outside the scope.